Cisco Final Project

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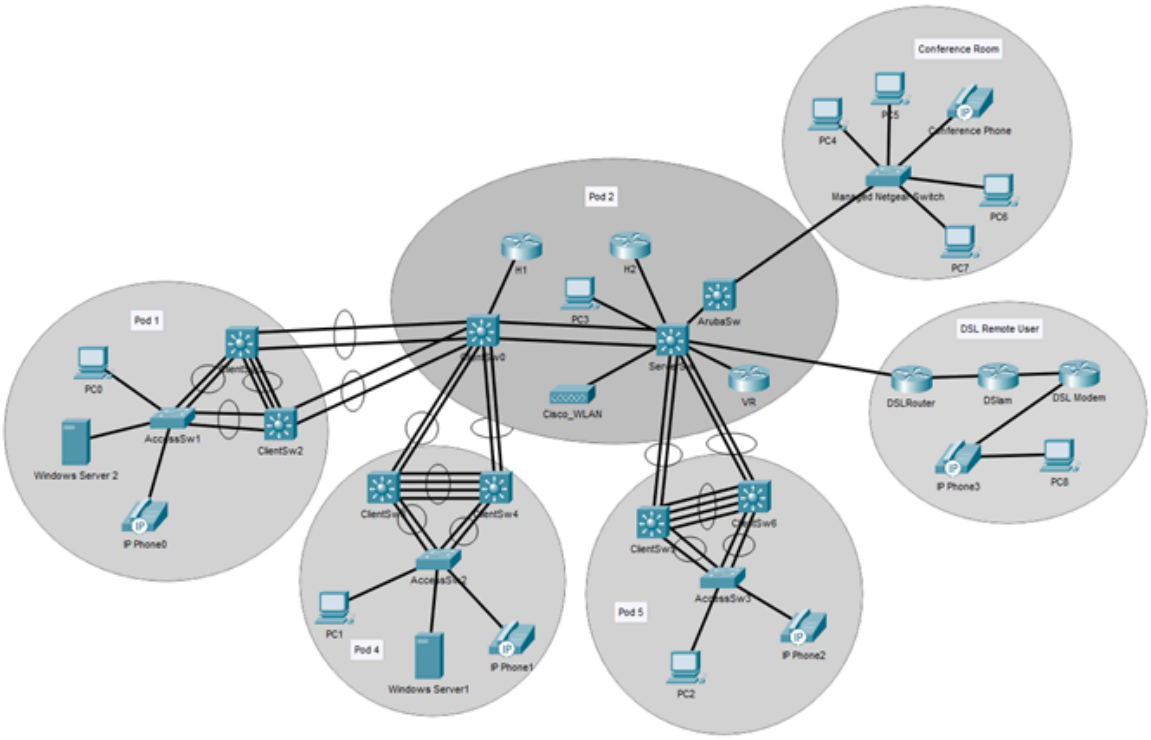

192.168.2.0/27

V99 - MANAGEMENT (192.168.2.0: .1-.30)

V10 - DATA (192.168.2.32: .33-.62)

V15 - VOICE (192.168.2.64: .65-.94)

V20 - DSL (192.168.2.96: .97-.126)

V25 - GUEST (192.168.2.128: .129-.158)

V30 - CONFERENCE ROOM (192.168.2.160: .

What’s Included:

- Windows Server with Active Directory and Group Policy
- HSRP for gateway redundancy between routers
- Redundant switch connections using Port Channels
- EIGRP routing configured across the network
- Static routes for internet access
- SSH access configured for secure management on routers
- NAT (Network Address Translation) for internet access
- DHCP services with multiple scopes for different VLANs
- VLAN segmentation for departments (VLANs 10, 15, 20, 25, 30, 99)
- Wireless Access Point configured with WPA2 security (Cisco_WLAN SSID)
- Network banner for unauthorized access warnings
- Port security applied on access ports
- Voice VLAN configuration for IP phones
- Inter-VLAN routing using router-on-a-stick (sub-interfaces)
- RIP and EIGRP redistribution
- VTP client configuration for switch VLAN management
- IP NAT with overload for internet sharing
- Device hostname and domain name configuration
- Encrypted passwords and service password encryption
- Static IP addressing for server infrastructure
- Basic telephony setup with ephone and dial-peer configuration
- IP default gateways configured for devices

H1

```
no ip domain lookup
ip domain name lab.local
banner MOTD #Unauthorized Access Is
Strictly Prohibited#
service password-encryprion
enable secret cisco
hostname H1
!
interface GigabitEthernet0/0/0
 description Link to Port 48 on ClientSw1
 ip address 192.168.2.14 255.255.255.224
 ip nat inside
 standby 1 ip 192.168.2.13
 standby 1 priority 105
 standby 1 preempt
 negotiation auto
!
interface GigabitEthernet0/0/1
 description Link to Internet
 ip address 192.168.1.152 255.255.255.0
 ip nat outside
 negotiation auto
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
 network 192.168.2.96 0.0.0.31
 network 192.168.2.128 0.0.0.31
 network 192.168.2.160 0.0.0.31
 network 192.168.2.224 0.0.0.31
!
ip route 0.0.0.0 0.0.0.0 192.168.1.1
!
crypto key generate rsa 2048
!
ip ssh version 2
ip ssh pubkey-chain
  username admin
!
ip nat inside source list 1 interface
GigabitEthernet0/0/1 overload
!
ip access-list standard 1
 10 permit any
!
line vty 0 15
 transport input ssh
!
line con 0
 password cisco
 login
end
```

H2

```
no ip domain lookup
ip domain name lab.local
banner MOTD #Unauthorized Access Is
Strictly Prohibited#
service password-encryprion
enable secret cisco
hostname H2
!
interface GigabitEthernet0/0/0
 description Link to ServSw on G1/0/47
 ip address 192.168.2.15 255.255.255.224
 ip nat inside
 standby 1 ip 192.168.2.13
 standby 1 preempt
 negotiation auto
!
interface GigabitEthernet0/0/1
 description Link To Internet
 ip address 192.168.1.153 255.255.255.0
 ip nat outside
 negotiation auto!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
 network 192.168.2.96 0.0.0.31
 network 192.168.2.128 0.0.0.31
 network 192.168.2.160 0.0.0.31
 network 192.168.2.224 0.0.0.31
!
ip route 0.0.0.0 0.0.0.0 192.168.1.1
!
crypto key generate rsa 2048
!
ip ssh version 2
ip ssh pubkey-chain
  username admin
!
ip nat inside source list 1 interface
GigabitEthernet0/0/1 overload
!
ip access-list standard 1
 10 permit any
!
line vty 0 15
```

```
 transport input ssh
!
line con 0
 password cisco
 login
end
```

Vlan Router Main

```
no ip domain lookup
hostname VR1
banner MOTD #Unauthorized Access Is
Strictly Prohibited#
service password-encryprion
enable secret cisco
!
ip dhcp excluded-address 192.168.2.33
192.168.2.43
ip dhcp excluded-address 192.168.2.65
192.168.2.73
ip dhcp excluded-address 192.168.2.129
192.168.2.135
ip dhcp excluded-address 192.168.2.225
192.168.2.230
!
ip dhcp pool local10
 network 192.168.2.32 255.255.255.224
 default-router 192.168.2.33
 dns-server 192.168.2.60 192.168.2.61
!
ip dhcp pool local15
 network 192.168.2.64 255.255.255.224
 default-router 192.168.2.65
255.255.255.224
 dns-server 8.8.8.8
 option 150 ip 192.168.1.210
!
ip dhcp pool local25
 network 192.168.2.128 255.255.255.224
 default-router 192.168.2.129
 dns-server 8.8.8.8
!
ip dhcp pool local30
 network 192.168.2.224 255.255.255.224
 default-router 192.168.2.225
 dns-server 8.8.8.8
 option 150 ip 192.168.1.210
!
interface GigabitEthernet0/0
 no shut
 description link to Port 48 on ServSw
!
interface GigabitEthernet0/0.10
 description Sub-Interface for Vlan 10
 encapsulation dot1Q 10
 ip address 192.168.2.33 255.255.255.224
!
interface GigabitEthernet0/0.15
 description Sub-interface for Vlan 15
 encapsulation dot1Q 15
 ip address 192.168.2.65 255.255.255.224
!
interface GigabitEthernet0/0.20
 description Sub-interface for Vlan 20
 encapsulation dot1Q 20
 ip address 192.168.2.97 255.255.255.224
!
Interface GigabitEthernet0/0.25
 description Sub-Interface for Vlan 25 Guest
 encapsulation dot1Q 25
 ip address 192.168.2.129 255.255.255.224
!
Interface GigabitEthernet0/0.30
 description Sub-Interface for Vlan 30
Conference
 encapsulation dot1Q 30
 ip address 192.168.2.225 255.255.255.224
!
interface GigabitEthernet0/0.99
 description Sub-Interface for Vlan 99
 encapsulation dot1Q 99 native
 ip address 192.168.2.1 255.255.255.224
!
router eigrp 1
 network 192.168.2.0
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
 network 192.168.2.96 0.0.0.31
 network 192.168.2.128 0.0.0.31
 network 192.168.2.160 0.0.0.31
 network 192.168.2.196 0.0.0.31
 network 192.168.2.224 0.0.0.31
!
ip route 0.0.0.0 0.0.0.0 192.168.2.13
!
crypto key generate rsa general-keys
modulus 2048
!
ip ssh version 2
ip ssh pubkey-chain
  username admin
!
```

```
line vty 0 15
 login
 transport input ssh
!
line con 0
 password cisco
 login
!
end
```

Wireless Access Point

```
hostname WAP
!
dot11 ssid Cisco_WLAN
 no shut
 authentication open
 authentication key-management wpa
 guest-mode
 wpa-psk ascii Cisco123
!
bridge irb
!
!
interface Dot11Radio0
 encryption mode ciphers aes-ccm
 !
 ssid Cisco_WLAN
interface BVI1
 noshut
 ip address 192.168.2.131 255.255.255.224
!
ip default-gateway 192.168.2.129
!
end
```

ServerSw (Pod 2):

```
hostname ServSw
!
ip routing
!
no ip domain-lookup
!
interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel12
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel13
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/1
 description Link To Client1 at Pod2 on
g1/0/1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/2
 description Link To Client1 at Pod2 on
g1/0/2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/3
 description Link To Client5 at Pod5 on
g1/0/7
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 12 mode desirable
!
interface GigabitEthernet1/0/4
 description Link To Client5 at Pod5 on
g1/0/8
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 12 mode desirable
!
interface GigabitEthernet1/0/5
 description Link To Client6 at Pod5 on
g1/0/9
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 13 mode desirable
!
interface GigabitEthernet1/0/6
```

```
 description Link To Client6 at Pod5 on
g1/0/10
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 13 mode desirable
!
interface GigabitEthernet1/0/20
 switchport access vlan 10
 switchport mode access
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky
00be.438a.bbc9
 switchport port-security mac-address sticky
0800.2740.8816
 switchport port-security
!
interface GigabitEthernet1/0/44
 description Link to ArubaSW on G1/44
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/45
 description Link to Guest AP
 switchport access vlan 25
 switchport mode access
!
interface GigabitEthernet1/0/46
 description Link to Dslam on g0/
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/47
 description Link to H2 on g0/0/0
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/48
 description Link to Vlan Router on g0/0
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan10
 ip address 192.168.2.40 255.255.255.224
!
interface Vlan15
 ip address 192.168.2.72 255.255.255.224
!
interface Vlan20
 ip address 192.168.2.104 255.255.255.224
!
interface Vlan25
 ip address 192.168.2.130 255.255.255.224
!
interface Vlan30
 ip address 192.168.2.226 255.255.255.224
!
interface Vlan99
 ip address 192.168.2.11 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
 network 192.168.2.96 0.0.0.31
 network 192.168.2.128 0.0.0.31
 network 192.168.2.160 0.0.0.31
 network 192.168.2.192 0.0.0.31
 network 192.168.2.224 0.0.0.31
 redistribute rip metric 100000 100 255 1
1500
 eigrp stub connected summary
!
router rip
 version 2
 redistribute eigrp 1 metric 3
```

ClientSw0 (Pod 2):

```
hostname Pod2ClientSwitch
ip routing
!
no ip domain-lookup
!
interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel3
 switchport trunk encapsulation dot1q
```

```
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel7
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel8
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/1
 description Link to Server Switch port 1/0/1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/2
 description Link to Server Switch port 1/0/2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/3
 description Link to Pod1 CL1 Port 5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface GigabitEthernet1/0/4
 description Link to Pod1 CL1 Port 6
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface GigabitEthernet1/0/5
 description Link to Pod1 CL2 Port 5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode desirable
!
interface GigabitEthernet1/0/6
 description Link to Pod1 CL2 Port 6
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode desirable
!
interface GigabitEthernet1/0/7
 description Link to CL3 Port 23
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 7 mode desirable
!
interface GigabitEthernet1/0/8
 description Link to CL3 Port 24
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 7 mode desirable
!
interface GigabitEthernet1/0/9
 description Link to CL4 Port 5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 8 mode desirable
!
interface GigabitEthernet1/0/10
 description Link to CL4 Port 6
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 8 mode desirable
!
interface GigabitEthernet1/0/11
 switchport trunk encapsulation dot1q
 switchport mode trunk
 shutdown
!
interface GigabitEthernet1/0/48
 description Link to H1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan10
 ip address 192.168.2.41 255.255.255.224
!
interface Vlan15
 ip address 192.168.2.73 255.255.255.224
!
interface Vlan20
 ip address 192.168.2.105 255.255.255.224
```

```
!
interface Vlan99
 ip address 192.168.2.12 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 255.255.255.224
 network 192.168.2.32 255.255.255.224
 network 192.168.2.64 255.255.255.224
 network 192.168.2.96 255.255.255.224
!
ip classless
ip route 0.0.0.0 0.0.0.0 192.168.2.13
!
end
```

ClientSw1 (Pod 1):

```
hostname CS1
!
ip routing
!
vtp mode client
vtp domain lab.local
!
interface Port-channel2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel6
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/1
 description Connection to CS2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 6 mode desirable
!
interface GigabitEthernet1/0/2
 description Connection to CS2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 6 mode desirable
!
interface GigabitEthernet1/0/3
 description Connection to CS2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 6 mode desirable
!
interface GigabitEthernet1/0/4
 description Connection to CS2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 6 mode desirable
!
interface GigabitEthernet1/0/5
 description Connection to Pod2 Client
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface GigabitEthernet1/0/6
 description Connection to Pod2 Client
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface GigabitEthernet1/0/23
 description Connection to AS1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 5 mode desirable
!
interface GigabitEthernet1/0/24
 description Connection to AS1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 5 mode desirable
!
interface Vlan10
 ip address 192.168.2.34 255.255.255.224
!
interface Vlan15
 ip address 192.168.2.66 255.255.255.224
!
interface Vlan20
```

```
 ip address 192.168.2.98 255.255.255.224
!
interface Vlan99
 ip address 192.168.2.2 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
 network 192.168.2.96 0.0.0.31
 eigrp stub connected summary
!
end
```

ClientSw2 (Pod 1):

```
hostname CS2
!
ip routing
!
vtp mode client
!
interface Port-channel3
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel6
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet2/0/1
 description Connection to CS1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 6 mode desirable
!
interface FastEthernet2/0/2
 description Connection to CS1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 6 mode desirable
!
interface FastEthernet2/0/3
 description Connection to CS1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 6 mode desirable
!
interface FastEthernet2/0/4
 description Connection to CS1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 6 mode desirable
!
interface FastEthernet2/0/5
 description Connection to Pod2 Client
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode desirable
!
interface FastEthernet2/0/6
 description Connection to Pod2 Client
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 3 mode desirable
!
interface FastEthernet2/0/21
 description Connection to As1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 4 mode desirable
!
interface FastEthernet2/0/22
 description Connection to As1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 4 mode desirable
!
interface Vlan10
 ip address 192.168.2.35 255.255.255.224
!
interface Vlan15
 ip address 192.168.2.67 255.255.255.224
!
interface Vlan20
 ip address 192.168.2.99 255.255.255.224
!
```

```
interface Vlan99
 ip address 192.168.2.3 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
 network 192.168.2.96 0.0.0.31
 auto-summary
!
end
```

AS1 (Pod 1):

```
hostname AS1
!
interface Port-channel4
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel5
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/1
 switchport access vlan 99
 switchport mode access
!
interface GigabitEthernet1/0/9
 switchport access vlan 10
 switchport mode access
 switchport port-security maximum 2
!
interface GigabitEthernet1/0/10
 switchport access vlan 10
 switchport mode access
!
interface GigabitEthernet1/0/20
 switchport access vlan 10
 switchport mode access
!
interface GigabitEthernet1/0/21
 description Connection to Client 2
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 4 mode desirable
!
interface GigabitEthernet1/0/22
 description Connection to Client 2
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 4 mode desirable
!
interface GigabitEthernet1/0/23
 description Connection to Client 1
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 5 mode desirable
!
interface GigabitEthernet1/0/24
 description Connection to Client 1
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 5 mode desirable
!
interface Vlan10
 ip address 192.168.2.42 255.255.255.224
!
interface Vlan15
 ip address 192.168.2.74 255.255.255.224
!
interface Vlan99
 ip address 192.168.2.4 255.255.255.224
```

DSLAMRouter (Pod 3):

```
hostname DSLAMROUTER
!
interface FastEthernet0/0
 description connects to port 46 on server
switch
 ip address 192.168.2.16 255.255.255.224
 duplex auto
 speed auto
!
interface ATM0/IMA0.1 point-to-point
 ip address 192.168.2.193 255.255.255.224
 no atm enable-ilmi-trap
 pvc 1/101
!
interface Vlan20
 ip address 192.168.2.116 255.255.255.224
!
interface Vlan 99
 ip address 192.168.2.15 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.96 0.0.0.31
!
ip route 0.0.0.0 0.0.0.0 192.168.2.13
!
dial-peer voice 1003 voip
 destination-pattern 100.
```

```
 session target ipv4:192.168.1.210
!
telephony-service
 max-ephones 4
 max-dn 4
 ip source-address 192.168.2.193 port 2000
 auto assign 1 to 10
 auto assign 1 to 4
 max-conferences 8 gain -6
 transfer-system full-consult
 create cnf-files version-stamp Jan 01 2002
00:00:00
!
!
ephone-dn  1
 number 4000
!
ephone-dn  2
 number 4001
!
ephone  1
 device-security-mode none
 mac-address 0016.46BD.93E3
 type 7940
 button  1:1
```

DSLModem (Pod 3):

```
hostname DSLAMROUTER
!
interface FastEthernet0/0
 description connects to port 46 on server
switch
 ip address 192.168.2.16 255.255.255.224
 duplex auto
 speed auto
!
interface ATM0/IMA0.1 point-to-point
 ip address 192.168.2.193 255.255.255.224
 no atm enable-ilmi-trap
 pvc 1/101
!
interface Vlan20
 ip address 192.168.2.116 255.255.255.224
!
interface Vlan 99
 ip address 192.168.2.15 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.96 0.0.0.31
!
ip route 0.0.0.0 0.0.0.0 192.168.2.13
!
dial-peer voice 1003 voip
 destination-pattern 100.
 session target ipv4:192.168.1.210
!
telephony-service
 max-ephones 4
 max-dn 4
 ip source-address 192.168.2.193 port 2000
 auto assign 1 to 10
 auto assign 1 to 4
 max-conferences 8 gain -6
 transfer-system full-consult
 create cnf-files version-stamp Jan 01 2002
00:00:00
!
!
ephone-dn  1
 number 4000
!
ephone-dn  2
 number 4001
!
ephone  1
 device-security-mode none
 mac-address 0016.46BD.93E3
 type 7940
 button  1:1
```

ClientSw3 (Pod 3):

```
hostname C3Pod4
!
ip routing
!
interface Port-channel7
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel9
 switchport trunk native vlan 99
 switchport mode dynamic desirable
!
interface Port-channel10
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
```

```
interface Port-channel11
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel12
 switchport trunk native vlan 99
 switchport mode dynamic desirable
!
interface FastEthernet0/1
 description port channel 11 to C4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 11 mode desirable
!
interface FastEthernet0/2
 description port channel 11 to C4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 11 mode desirable
!
interface FastEthernet0/3
 description port channel 11 to C4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 11 mode desirable
!
interface FastEthernet0/4
 description port channel 11 to C4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 11 mode desirable
!
interface FastEthernet0/5
 description port channel 10 to A2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 10 mode desirable
!
interface FastEthernet0/6
 description port channel 10 to A2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 10 mode desirable
!
interface FastEthernet0/23
 description port channel 7 to main switch
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 7 mode desirable
!
interface FastEthernet0/24
 description port channel 7 to main switch
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 7 mode desirable
!
interface Vlan10
 ip address 192.168.2.36 255.255.255.224
!
interface Vlan15
 description voice
 ip address 192.168.2.68 255.255.255.224
!
interface Vlan20
 ip address 192.168.2.100 255.255.255.224
!
interface Vlan99
 ip address 192.168.2.5 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
 network 192.168.2.96 0.0.0.31
```

ClientSw4 (Pod 4):

```
hostname C4Pod4
!
ip routing
!
interface Port-channel8
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel9
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel10
 switchport trunk native vlan 99
```

```
!
interface Port-channel11
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet1/0/1
 description port channel 11 to C3Pod4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 11 mode desirable
!
interface FastEthernet1/0/2
 description port channel 11 to C3Pod4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 11 mode desirable
!
interface FastEthernet1/0/3
 description port channel 11 to C3Pod4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 11 mode desirable
!
interface FastEthernet1/0/4
 description port channel 11 to C3Pod4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 11 mode desirable
!
interface FastEthernet1/0/5
 description port channel 8 to
Pod2ClientSwitch
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 8 mode desirable
!
interface FastEthernet1/0/6
 description port channel 8 to
Pod2ClientSwitch
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 8 mode desirable
!
interface FastEthernet1/0/7
 description port channel 9 to A2Pod4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 9 mode desirable
!
interface FastEthernet1/0/8
 description port channel 9 to A2Pod4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 9 mode desirable
!
interface Vlan10
 ip address 192.168.2.37 255.255.255.224
!
interface Vlan15
 ip address 192.168.2.69 255.255.255.224
!
interface Vlan20
 ip address 192.168.2.101 255.255.255.224
!
interface Vlan99
 ip address 192.168.2.6 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
 network 192.168.2.96 0.0.0.31
```

AccessSw2 (Pod 4):

```
hostname AccessSw2
!
interface Port-channel1
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel2
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/5
 description link to C3Pod4
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/6
 description Link to C3Pod4
```

```
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/7
 description Link to C4Pod4
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface GigabitEthernet1/0/8
 description Link to C4Pod4
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 2 mode desirable
!
interface GigabitEthernet1/0/10
 switchport access vlan 10
 switchport mode access
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky
00be.438a.bb8d
 switchport port-security mac-address sticky
0800.27f3.0374
 switchport port-security
!
interface GigabitEthernet1/0/11
 switchport access vlan 10
 switchport mode access
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky
00be.438a.b9e2
 switchport port-security
!
interface GigabitEthernet1/0/12
 switchport access vlan 10
 switchport mode access
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 switchport port-security
!
interface Vlan99
 ip address 192.168.2.7 255.255.255.224
```

ClientSw5 (Pod 5):

```
hostname Core5
!
interface Port-channel12
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel15
!
interface Port-channel16
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 12 mode desirable
!
interface GigabitEthernet0/2
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 12 mode desirable
!
interface GigabitEthernet0/3
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 16 mode desirable
!
interface GigabitEthernet0/4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 16 mode desirable
!
interface GigabitEthernet0/5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 16 mode desirable
!
interface GigabitEthernet0/6
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 16 mode desirable
!
interface GigabitEthernet0/7
```

```
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 12 mode desirable
!
interface GigabitEthernet0/8
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 12 mode desirable
!
interface GigabitEthernet0/9
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/10
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/11
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/12
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/13
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/17
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/18
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet0/23
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 15 mode desirable
!
interface GigabitEthernet0/24
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 15 mode desirable
!
interface Vlan10
 ip address 192.168.2.38 255.255.255.224
!
interface Vlan15
 ip address 192.168.2.70 255.255.255.224
!
interface Vlan20
 ip address 192.168.2.102 255.255.255.224
!
interface Vlan99
 ip address 192.168.2.8 255.255.255.224
!
router eigrp 1
 network 192.168.2.0 0.0.0.31
 network 192.168.2.32 0.0.0.31
 network 192.168.2.64 0.0.0.31
```

ClientSw6 (Pod 5):

```
hostname client6
!
interface Port-channel13
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Port-channel14
 switchport trunk native vlan 99
!
interface Port-channel16
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
!
interface FastEthernet0/1
 description connects to pgbT ba0/3 on
client 5
!
interface FastEthernet0/2
 description connects to port fa0/2 on client
5
!
interface FastEthernet0/3
```

```
 description connects to port fa0/3 on client
5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 switchport nonegotiate
 channel-group 16 mode desirable
!
interface FastEthernet0/4
 description connects to port fa0/4 on client
5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 16 mode desirable
!
interface FastEthernet0/5
 description connects to port fa0/5 on client
5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 16 mode desirable
!
interface FastEthernet0/6
 description connects to port fa0/6 on client
5
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 16 mode desirable
!
interface FastEthernet0/7
 description connects to port fa0/5 on
ServerSWRight
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 13 mode desirable
!
interface FastEthernet0/8
 description connects to port fa0/6 on
ServerSWRight
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 13 mode desirable
!
interface FastEthernet0/21
 description conects to fa0/21 on access
switch 3
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 14 mode desirable
!
interface FastEthernet0/22
 description conects to fa0/22 on access
switch 3
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 99
 switchport mode trunk
 channel-group 14 mode desirable
!
interface Vlan10
 ip address 192.168.2.39 255.255.255.224
!
interface Vlan15
 ip address 192.168.2.71 255.255.255.224
!
interface Vlan20
 ip address 192.168.2.103 255.255.255.224
!
interface Vlan99
 ip address 192.168.2.9 255.255.255.224
```

AccessSw3 (Pod 5):

```
hostname AccessSwitch3
!
interface GigabitEthernet1/0/1
 switchport access vlan 10
 switchport mode access
!
interface GigabitEthernet1/0/2
 switchport access vlan 10
 switchport mode access
!
interface GigabitEthernet1/0/3
 switchport access vlan 10
 switchport mode access
!
interface GigabitEthernet1/0/4
 switchport access vlan 10
 switchport mode access
!
interface GigabitEthernet1/0/5
 switchport access vlan 10
 switchport mode access
!
interface GigabitEthernet1/0/6
 switchport trunk native vlan 99
```

```
 switchport mode trunk
!
interface GigabitEthernet1/0/7
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/8
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/9
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/10
 description connects to pc1
 switchport access vlan 10
 switchport trunk native vlan 99
 switchport mode access
!
interface GigabitEthernet1/0/11
 switchport trunk native vlan 99
 switchport mode trunk
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/18
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/19
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/20
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/21
 description connects to access switch 6
port 21
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/22
 description connects to acces switch 6 port
22
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/12
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/13
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/14
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/15
 switchport access vlan 10
 switchport trunk native vlan 99
 switchport mode access
!
interface GigabitEthernet1/0/16
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/17
 description connects to access switch 5
port 23
 switchport trunk native vlan 99
 switchport mode trunk
!
interface GigabitEthernet1/0/24
 description connects to access switch 5
port 24
 switchport trunk native vlan 99
 switchport mode trunk
!
interface Vlan99
 ip address 192.168.2.10 255.255.255.224
```